

AIRSPEED: An Open-source Universal Data Production Platform for Embodied Artificial Intelligence

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ABSTRACT

Embodied artificial intelligence development hinges on effective data acquisition, yet it faces challenges such as high costs, limited training scenarios, and a lack of standards for dataset. To overcome these, we introduce AIRSPEED, an open-source platform for embodied artificial intelligence data production. AIRSPEED features hardware-software decoupling to reduce costs, supports multiple devices for diverse data collection, interfaces with various simulation platforms for synthetic data generation, and automates dataset construction. It comprises three services: data collection for real-world data, data generation for synthetic data, and dataset construction for organizing data pyramid structures. AIRSPEED aims to enhance data production efficiency and quality, thereby improving embodied artificial intelligence model performance. Our experiment confirmed the effectiveness of AIRSPEED in improving data production efficiency. All software are open-sourced and can be found on this website: <https://AIRSPEED.ai>.

KEYWORDS

Embodied Artificial Intelligence, Robotics, Data collection, Dataset

1 INTRODUCTION

The Autonomy Economy represents a transformative phase in our society, driven by the integration of autonomous machines such as autonomous vehicles, delivery robots, drones, and more into the provision of goods and services. Central to this revolution is Embodied Artificial Intelligence (EAI), the intelligence backbone enabling these diverse autonomous systems [1]. EAI involves integrating artificial intelligence into physical entities, such as robots, enabling them to perceive, learn from, and interact dynamically with their surroundings [2, 3].

Existing research indicates that the development of EAI still conforms to the scaling laws in the field of artificial intelligence [4]. This means that more high-quality data can produce higher performance EAI models. However, there are still many difficulties in the large-scale collection and generation of EAI data. From a purely software perspective, on the one hand, data collection software for different robots requires specialized customization, making it difficult to unify data collection standards. On the other hand, different robots and algorithms are compatible with various simulation platforms, making the integration and utilization of simulation data challenging. In addition, the current dataset construction method relies heavily on manual operations, which seriously reduces the production efficiency of data. These difficulties pose barriers to efficient data production.

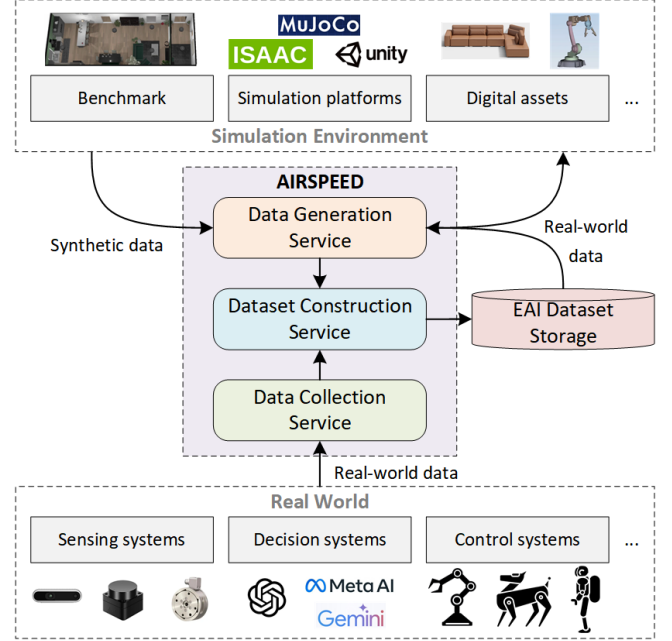


Figure 1: AIRSPEED architecture.

To address these EAI data production bottleneck, we propose AIRSPEED, an open-source universal data production platform for EAI. Fig. 1 provides a system overview of AIRSPEED architecture. AIRSPEED has embedded three services: data collection service, data generation service, and dataset construction service. AIRSPEED collects various types of real-world data through its data collection service on one hand, and on the other hand, it gathers synthetic data from multiple simulation environments through its data generation service. The dataset construction service can automatically build a pyramid-structured EAI dataset, and its format is compatible with the conversion to other dataset formats. AIRSPEED can coordinate the collection and generation of various EAI data and is highly compatible with different types of robots, data collection methods, and robotic simulation platforms, providing flexible training data for various embodied intelligence technologies. Overall, AIRSPEED has brought three contributions as follows:

- **Universal data production platform:** AIRSPEED utilizes a unified robot interface and simulation interface to connect with various data collection hardware and simulation platforms, achieving decoupling of hardware and software, thereby reducing the development costs on the software side for data producers.

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- **Support multiple EAI data collection and generation methods:** AIRSPEED supports data collection based on teleoperation and demonstration, as well as data generation based on data synthesis and virtual teleoperation, offering the most comprehensive support for current technologies.
- **Automated dataset construction:** AIRSPEED supports the automated construction of pyramid-level datasets according to user configurations, which can meet the customized needs for various model training requirements.

2 SURVEY OF EAI DATA PRODUCTION

As shown in Fig. 2, the production of EAI data comes from two sources: data collection from the real world and data generation from the simulation environment. Each of these sources has given rise to a variety of technical solutions. Different technical solutions correspond to different data production methods, hence many scholars have conducted related research. Another key aspect of EAI data production is dataset construction, which directly determines the purpose of data usage. As a data production platform, AIRSPEED can support data collection, data generation, dataset construction, and cover most technical branches. In this section, we conduct a comprehensive survey of data production to summarize the current related work.

2.1 EAI Data Collection

EAI data collection has two methods: Teleoperation approach and demonstration approach.

Teleoperation, also known as telerobotics, refers to the remote control of robots or mechanical systems by human operators from a distance [5]. The term "tele-" implies long-distance operation, allowing an operator to manipulate a robot's actions from a remote location. Teleoperation approach can be specifically divided into four categories:

- **Wearable teleoperation.** Wearable teleoperation refers to a type of technology that allows human operators to remotely control robots through the use of wearable equipment. These devices capture the movements and gestures of the operator and translate them into control signals for the robot. The wearable equipment can include gloves [6], suits [7], or exoskeletons [8] that are equipped with sensors to track the operator's movements or provide haptic feedback.
- **Visual-based teleoperation.** Visual-based teleoperation refers to the process where an operator's movements are captured using visual sensing technology, such as cameras [6] [9], and these movements are then translated into control commands to maneuver a robot. This method allows for the intuitive control of robotic systems by directly mapping human actions to robotic movements, which is suitable for cost saving and low precision requirements.
- **Optical-inertial teleoperation.** Optical-inertial teleoperation is a sophisticated method that combines the use of optical motion capture systems with inertial measurement units (IMUs) to control a robot remotely. This approach leverages the strengths of both wearable and visual-based technologies to achieve more accurate, reliable, and continuous tracking

of the operator's movements, such as motion capture system [10], virtual reality-based teleoperation system [11] and other forms of integration [12].

- **Isomorphic teleoperation.** Isomorphic teleoperation refers to a control method where the teleoperator or the control system is designed to have a one-to-one correspondence, or isomorphism, between the master (operator's) device and the slave (robotic) device, such as ALOHA [13] and Mobile ALOHA [14]. This means that the kinematic structure of the master device mirrors that of the slave device, allowing for a more intuitive and direct control of the robot's movements.

Demonstration refers to a pedagogical approach where a human operator executes a task or a sequence of tasks, which are then meticulously recorded to instruct a robot in performing analogous tasks [15]. This method is integral to robot learning from demonstration (LfD), a subfield of robotics that focuses on algorithms enabling robots to acquire behaviors by observing examples provided by a teacher. Demonstration approach can be divided into three categories:

- **Drag teaching.** Drag teaching refers to a method where an operator physically manipulates the robot's joints or end-effector to desired positions, which are then recorded and stored as a sequence of movements that the robot can later reproduce [16]. This technique is particularly useful for programming robots in a more intuitive and interactive manner, especially for tasks that require precise positioning or complex motions.
- **Wearable demonstration.** Unlike wearable teleoperation, wearable demonstration refers to the process where humans perform tasks directly through wearable devices, and the data recorded by these devices can be used directly for robotic learning without the conversion from human movements to robot movements. For example, using a wearable claw for data collection as UMI [17].
- **Human video learning.** Human video learning refers to the process where robots learn from videos of humans performing various tasks, such as GRH [18], MimicPlay [19] and GR-2 [20]. This method leverages the wealth of human demonstration data available in video formats, which can be more cost-effective and easier to collect than expert demonstrations with real robots.

Facing a variety of data collection methods, many scholars attempt to develop universal collection techniques. For instance, GELLO [21] aims to lower the barrier to collecting large and high-quality human demonstration data by proposing a general framework for building low-cost and intuitive teleoperation systems for robotic manipulation. PATO [22] automates part of the demonstration collection process using a learned assistive policy, enabling a single operator to control multiple robots in parallel for scalable robotic data collection. However, these technologies still only cover some collection methods and have limited universality. Recently, DART [23] has provided a more convenient way for data collection, which is a teleoperation platform designed for crowdsourcing that reimagines robotic data collection by leveraging cloud based simulation and augmented reality (AR). DART can perform various types of teleoperation for data collection on a unified platform,

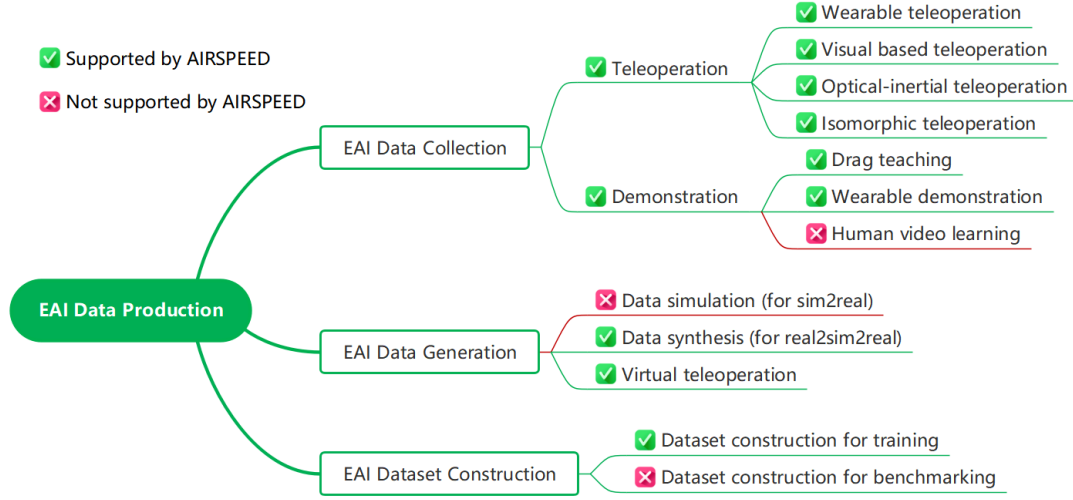


Figure 2: Production methods of EAI data and AIRSPEED’s support for them.

indicating that the universality of the data collection platform has gradually gained attention from scholars.

2.2 EAI Data Generation

EAI data generation utilizes virtual simulation environments to create a wide range of scenarios and conditions that a robot may encounter. It provides a controlled, safe, and cost-effective way to generate the extensive data needed for training and testing robotic systems. EAI data generation has three methods:

- **Data simulation.** Data simulation refers to data generated based on various simulation platforms, such as PyBullet [24], Isaac Sim [25], MuJoCo [26], etc. Data simulation only uses manually created digital assets for simulation tasks, which often creates a gap between them and the real world.
- **Data synthesis.** Data synthesis refers to the process of importing control model or real-world data into simulation platforms and then generating synthetic data through related algorithms. For example, the ACT algorithm can manipulate robots in a simulated environment to generate more synthetic data through a trained control model [13], while ACDC converts objects and scenes in RGB images into digital cousins and further generates synthetic data [27].
- **Virtual teleoperation.** Virtual teleoperation refers to the generation of robot data by sending remote control commands to the simulation platform via external devices such as keyboard, remote control handle, and motion capture device [10]. Through communication interfaces, the vast majority of simulation platforms or algorithms support virtual teleoperation.

Under these three methods, many scholars and companies have proposed corresponding data generation frameworks. For example, BestMan [28], based on PyBullet, has developed A modular mobile manipulator platform for EAI with unified simulation-hardware APIs for simulation data generation. Mimicgen [29] proposed A data generation system for scalable robot learning using human

demonstrations, which can generate a large amount of synthetic data from a small amount of real-world data. VitRob Pipeline [30] proposed a seamless teleoperation pipeline for advanced virtual reality, which can generate robot data for crop harvesting through virtual teleoperation. However, their designs consider more the convenience of model training rather than the universality of large-scale data generation.

2.3 EAI Dataset Construction

The construction of EAI datasets can be divided into two categories based on their purpose: for training and for benchmarking. The former is mostly composed of real-world data, while the latter is largely built on simulation platforms, thus for the latter, data generation and dataset construction can be considered as a single entity.

- **Dataset construction for training.** Typical datasets in this category include Open X-Embodiment [31] and ARIO [32]. Open X-Embodiment is a collaboration of 21 institutions collected an embodied dataset from 22 different robots, showcasing 527 skills and 160,266 tasks. ARIO (All Robots In One) is a dataset that comprising approximately 3 million episodes collected from 258 series and 321,064 tasks. The construction of these datasets relies on experts, is costly and time-consuming, and lacks a unified dataset standard.
- **Dataset construction for benchmarking.** Typical datasets in this category include Habitat3.0 [33], EmbodiedCity [34], MetaUrban [35], and GRUtopia [36]. Since this paper focuses more on the collection of real data, this type of dataset construction is not the focus of this paper.

3 AIRSPEED SYSTEM DESIGN

The design of AIRSPEED is aimed at satisfying three purposes:

- **Efficient large-scale real-world data collection.** EAI systems are typically equipped with a diverse array of sensors, such as cameras and LiDARs, which generate vast amounts

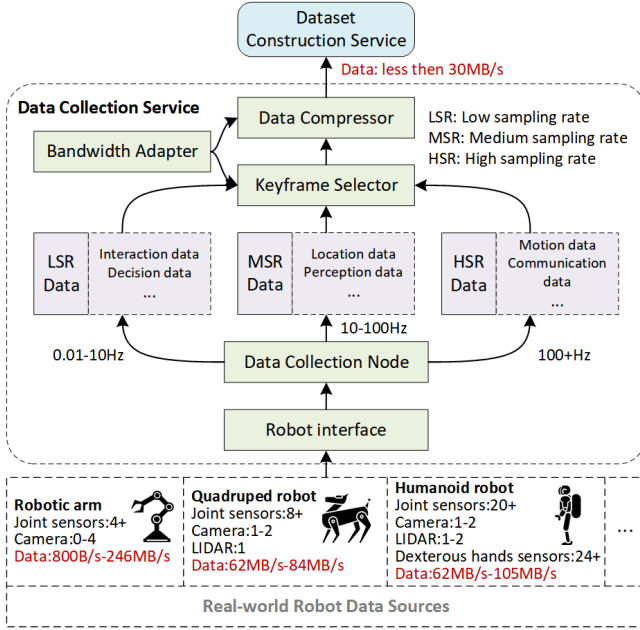


Figure 3: Data collection service of AIRSPEED

of data – ranging from tens to hundreds of megabytes per second. To ensure data integrity, low latency and high bandwidth in data transmission are crucial for AIRSPEED to minimize the time spent collecting massive data.

- **Convenient data circulation in simulation environments.** The gap between simulation data and real-world data poses a significant challenge in the advancement of embodied intelligence. AIRSPEED is designed to offer flexible simulation environment configurations, triggering mechanisms, and data alignment, all aimed at minimizing this gap as much as possible.
- **Automated construction of a pyramidal structure dataset.** In the field of embodied intelligence, different technological approaches imply varying data collection needs. AIRSPEED is capable of automatically constructing datasets in a pyramid structure to accommodate various data formats and facilitate their interconversion.

To this end, AIRSPEED has embedded three services: data collection service, data generation service, and dataset construction service, providing full-process customization for data collection, data generation, and dataset construction. As shown in Fig. 2, through these three services, AIRSPEED is capable of being compatible with the majority of data collection and generation methods, making it convenient for users to employ as a universal data production platform.

3.1 Data Collection Service

The design of AIRSPEED’s data collection service takes into account both compatibility and performance. First, the data collection service needs to be capable of being compatible with as many data sources, data modalities, and data formats as possible. Secondly, EAI

data collection must address four critical factors: data latency, data transmission bandwidth, data quality, and computing resources.

Data latency can cause time alignment errors between different modalities of data, thereby affecting data availability. Data transmission bandwidth including network bandwidth and the local storage buffer capacity. Insufficient data transmission bandwidth can exacerbate latency issues and even lead to data collection failures. Data quality refers to the degree of lossless data, and data collection services should strive to ensure the lossless nature of data as much as possible. Data latency, data transmission bandwidth, and data quality all need to achieve a compromise under limited computing resources.

The lower half of Fig. 3 illustrates the potential data traffic size from various data sources, ranging from 800B/s (if we collect motion data only) to 246MB/s depending on the type and number of sensors. When collecting data from multiple robots in parallel, EAI data influx can easily greatly exceeds the available data transmission bandwidth, resulting in data collection failures if not managed properly.

Therefore, the data collection service have one overarching design goal: balancing the above four factors in large-scale data collection. As shown in Fig. 3, we designed a highly compatible robot interface and a dynamic data collection mechanism by a data collection node, bandwidth adaptor, keyframe selector, and data compressor within the data collection service to achieve the optimal point while facing these trade-offs.

Robot Interface. The Robot Interface is designed for widespread compatibility with various robots’ data transmission interfaces, receiving data returned by the robot based on the data collection method. Users can determine the reception method of the data by writing configuration files, including which data to receive, the format of the data, the data frame rate, units, etc. Users only need to write configuration files to quickly receive robot data without programming.

Data Collection Node. Data collection node is a communication node designed based on a robot operating system such as ROS2 [37] and DORA [38]. It’s design goal is to collect as complete raw data as possible and perform data classification based on the data sampling rate. The data collection node will classify data into LSR (Low Sampling Rate) data (0.01-10Hz), MSR (Medium Sampling Rate) data (10-100Hz), and HSR (High Sampling Rate) data (100+Hz) according to the original frequency of the data, for subsequent data processing strategies.

Data of different frequencies has varying demands on the performance required for data transmission [39]. Low frequency implies that the data is not sensitive to transmission latency, allowing algorithms to allocate more computing resources to compress the data while ensuring data quality, in order to save transmission bandwidth. High frequency indicates that data is sensitive to transmission latency, requiring algorithms to ensure sufficient performance resources for high-speed data transmission. Medium frequency falls between the two, with most visual data typically in this range. Algorithms need to adjust resources for keyframe selection and data compression according to changes in data transmission bandwidth, to balance the contradiction between data latency and data quality.

Bandwidth Adapter. The task of the bandwidth adapter is to continuously monitor the current data transmission bandwidth and

dynamically adjust the strategies of the keyframe selector and data compressor to achieve the optimal data collection quality under current conditions.

The bandwidth adapter categorizes the usage of data transmission bandwidth into three classes: low data traffic, medium data traffic, and high data traffic. Low data traffic implies a high bandwidth surplus, where the keyframe selector and Data Compressor can reduce the consumption of performance resources or even remain idle, thereby reducing power consumption. High data traffic implies a low bandwidth surplus, requiring the keyframe selector and Data Compressor to prioritize obtaining performance resources to ensure the successful transmission of data. Medium data traffic is the most ideal state, which means that the keyframe selector and Data Compressor can maintain a low power state and have the potential to respond to changes in data traffic.

Keyframe Selector. The task of the keyframe selector is to select keyframes from the data according to the learning objectives and to remove redundant data frames in proportion. The deletion of redundant data frames will inevitably lead to a decrease in data quality and an increase in time alignment errors, so its removal ratio needs to be controlled according to the Bandwidth Adapter.

The detection of keyframes is the core algorithm for constructing a keyframe selector. In scenarios where visual sensory data accounts for more than 99.9% of all data, the detection of keyframes is essentially equivalent to keyframe detection of visual data [40]. Keyframe detection algorithms can be categorized into three types: heuristics-based methods, probabilistic-based methods, and learning-based methods [40]. Regardless of the method used, the premise is to ensure the quality of visual information to the greatest extent, as demonstrated in [41] and [42]. AIRSPEED keyframe selector is designed to be extensible so as to integrate various key frame selection algorithms, allowing users to select the most suitable algorithms. To ensure data quality, keyframe selection rate can be as low as 2% to still achieve similar training performance, a 50x reduction of data throughput.

Data Compressor. The task of the data compressor is to compress data according to the data transmission bandwidth. Data compression may be lossless or lossy, and lossy compression will inevitably lead to a decrease in data quality, so its compression strategy needs to be controlled according to the bandwidth adapter.

The data compressor supports a variety of compression methods. Taking images as an example, it supports lossless compression methods such as JPEG-LS, PNG, and WebP, and lossy compression methods including JPEG, JPEG-2000, and WebP. Some image compression methods use GPU based improvements to achieve better real-time performance [43]. We have achieved up to 50x compression ratio and compression throughput of 296 MB/s. Similar to the keyframe selection, AIRSPEED data compressor is designed to be extensible so that the users can select the most suitable algorithm for data compression.

In our design, after going through the whole data collection service, we managed to filter down data influx by up to 30x, thus successfully achieving real-time EAI data collection under different real-world conditions.

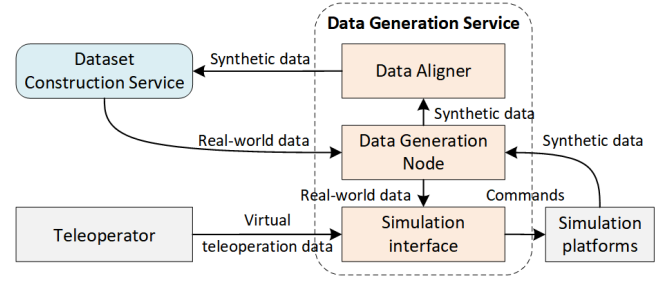


Figure 4: Data generation service of AIRSPEED.

3.2 Data Generation Service

The EAI data generation service in AIRSPEED serves two needs: facilitating convenient data transmission between real-world and simulation environment, and efficient data alignment between real-world data and synthetic data. As shown in Fig. 4, the former is implemented by the data generation node and the simulation interface, and the latter is realized by the data aligner.

Data Generation Node. The data generation Node has three functions: receiving real-world data from the data collection service, sending real-world data or virtual teleoperation data to the simulation interface, and receiving synthetic data generated by simulation platforms. Similar to the data collection node in the data collection service, it is also a communication node designed based on ROS2 or DORA. The data generation node supports communication with various simulation platforms, such as PyBullet [24], Isaac Sim [25], MuJoCo [26], etc..

Simulation Interface. Receiving synthetic data output from simulation platforms is a relatively simple task; however, integrating different types of real-world data and virtual teleoperation data into different simulation platforms is a cumbersome engineering challenge. To address this, we have developed a universal simulation interface within the data generation service to simplify the data integration process as much as possible. Similar to the robot interface, the simulation interface is designed for widespread compatibility with simulation platforms' data transmission interfaces. Users can determine the reception method of the data by writing configuration files.

Data Aligner. The alignment between real-world collected data and simulated data of robots is a critical aspect of EAI data production. It involves several key components that ensure the data from both sources can be effectively integrated and utilized for training. There is a data aligner integrated in the data generation service of AIRSPEED. It is responsible for temporal alignment, spatial alignment, and unification of physical units.

Temporal and spatial alignment ensures that data from different sources are synchronized in time and space. This is crucial for tasks that require precise coordination between sensory inputs and robotic actions. For instance, when training a robot to navigate an environment, the visual input from cameras must be accurately aligned with the robot's movements and the corresponding timestamps [44]. And the frame rate of the simulated data may not match the frame rate of the real sensor, so corresponding alignment algorithms are needed for calibration.

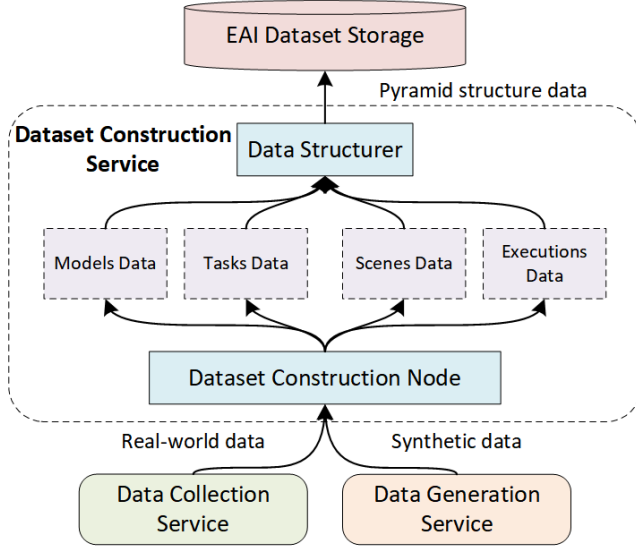


Figure 5: Dataset construction service of AIRSPEED.

The unification of physical units refers to the process of ensuring that all measurements, regardless of their source, are expressed in consistent units of time, distance, force, and other physical quantities. This standardization is essential for accurate data comparison and integration, allowing for a direct and meaningful correlation between real-world observations and their simulated counterparts.

3.3 Dataset Construction Service

The dataset construction service is used to structure collected and generated data according to user-defined rules to automatically build a pyramid-structured dataset. To this end, we have designed a dataset construction node and a data structurer.

Dataset Construction Node. The dataset construction node is not only a communication node but also responsible for categorizing data into different levels based on functionality: models data, scenes data, tasks data, and executions data. This data categorization is to facilitate the formation of a pyramid-structured dataset.

Models data refers to data related to the robot’s entity (in the real world or in simulation environments), such as robot model data, sensor model data, and simulation environment hardware and software model data.

Scenes data refers to data related to the internal and external parameter configurations of the robot, such as sensor calibration data, classification data of the scenes where the robot is located (indoor/outdoor, grassland/sand, day/night, etc.), and map data of the scenes where the robot is situated.

Tasks data refers to data related to the tasks executed by the robot, such as the instruction data received by the robot, the initial state of the robot, the type of skills used by the robot, and descriptive data of the objects being operated on.

Executions data refers to the data generated by the robot during the execution of tasks, such as the robot’s motion data (position, velocity, force, deflection angle, etc.), various sensor data, and decision-making process data.

AIRSPEED defaults to dividing data into the four levels mentioned above, yet users can still customize the category levels as they wish.

Data Structurer. The data structurer is used to organize the collected data according to the hierarchical structure of the pyramid; its structuring rules are not set in stone and are determined by user configuration. For example, users can design the models data as the top level based on different robot models, use scenes data as the second level based on different scenarios, tasks data as the third level based on different tasks, and executions data as the fourth level based on different execution instances. Alternatively, scenes data can be used as the top level based on different scenarios, models data as the second level, tasks data as the third level based on different tasks, and executions data as the fourth level based on different execution instances.

AIRSPEED defaults to building the pyramid structure dataset in the order of Models-Tasks-Scenes-Executions. Users can alter the pyramid structure by writing configuration files and can define the more detailed structure of each level with precision. This provides a flexible way for users to build custom datasets.

3.4 Design Advantages

Compared to other existing data production platforms, software, and devices, AIRSPEED has the following advantages:

- **Hardware-software decoupling.** AIRSPEED is a universal pure software platform whose data collection capabilities are independent of the teleoperation devices and robot models used in the data collection system. Users only need to write configuration files to use it, reducing the development and usage costs of the software part in the data collection system.
- **Fully automated process.** The data collection, data generation, and dataset construction processes in AIRSPEED are all automated and can be carried out in parallel. This not only eliminates the manpower and time consumed in data transfer and data organization but also makes full use of idle computing resources, improving resource utilization.

4 EXPERIMENTS

AIRSPEED’s universality is based on three key advantages: the universality of the platform backbone, support for a variety of data production methods, and superior performance. This section will provide a detailed experiments to these three aspects.

4.1 Universality of Platform Backbone

To realize the system architecture of AIRSPEED, a platform backbone is required to connect different components together and to host various services. Two robot operating systems, DORA [38] and ROS2 [37], became backbone candidates for AIRSPEED due to their high performance and popularity among various robotics workloads.

DORA is designed to support dataflow computing, which streamlines data in a dataflow fashion. Particularly, services in DORA are

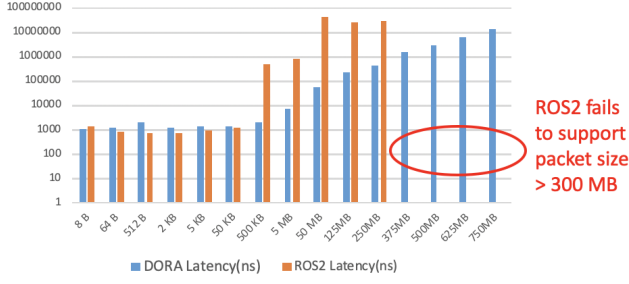


Figure 6: Platform backbone performance comparison between DORA and ROS2 of transmission latency

modeled as directed graphs, where data flows through streams and is processed by nodes, often resembling a sequence of connected nodes referred to as pipelines [45]. This architecture allows for efficient data processing and flexibility in application design.

Conversely, ROS2 provides a robust framework for robot software development, emphasizing modularity and reusability. It supports a wide range of hardware interfaces and boasts an extensive ecosystem of tools and libraries. However, it may not offer the same level of low-latency dataflow optimization as DORA.

To compare the performance of DORA and ROS2, we conducted an experiment measuring the latencies of transmitting packets across a wide range of sizes (from 8B to 750MB), emulating various forms of robotic data. The results of this experiment are summarized in Fig. 6. The X-axis shows different packet sizes, whereas Y-axis shows transmission latency in nanoseconds and is presented in log scale. For packets less than 50KB, ROS2 and DORA have similar performance, but for packets larger than 500KB, DORA significantly outperforms ROS2. One particularly interesting case is 5 MB packet, which is commonly used for sending image data and most popular among robotic workloads, the latencies on ROS2 are consistently around 100 times larger than the latencies on DORA, and we have repeated our experiments 10 times to verify the results.

AIRSPEED supports users in setting up DORA or ROS2 as the platform backbone. For simplicity, AIRSPEED defaults to using ROS2 as the platform backbone.

4.2 Data production experiments

As shown in Fig. 2, AIRSPEED supports a variety of data production methods. For each method, AIRSPEED has developed corresponding robot and simulation interfaces, and users only need to perform simple configurations to utilize them.

Among these methods, teleoperation is the most widely applied technology and also the most complex in terms of data production system design. Fig. ?? illustrates the AIRSPEED software architecture for teleoperation data production built on ROS2. It includes three types of modules: ROS2 node, hardware device, and software service. From the diagram, it can be seen that the robot interface, simulation interface, data collection service, and data generation service are all ROS2 communication nodes, which means they can be deployed on different computing devices. Moreover, the dataset construction service supports both local and cloud deployment, indicating that AIRSPEED can be a distributed data production

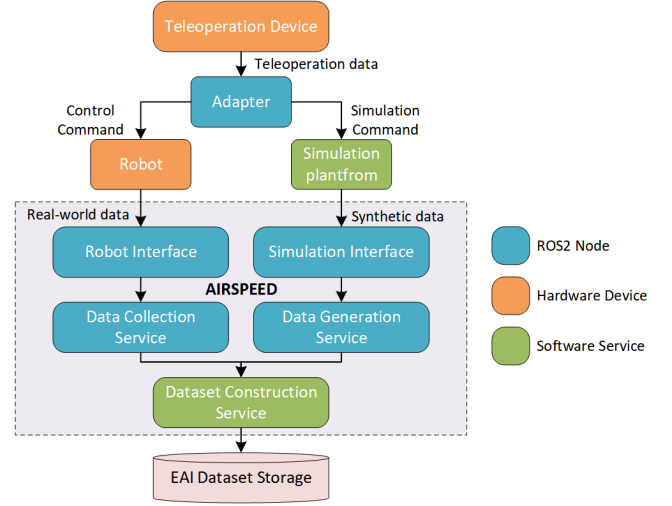


Figure 7: AIRSPEED's software architecture for teleoperation data production built on ROS2

system, offering users flexible deployment options. The adapter is also a ROS2 node, used for sending control commands to robots and simulation commands to the simulation platform; it is not part of AIRSPEED. However, in our open-source code, we provide a demo example to facilitate extensive teleoperation device adaptation by the open-source community.

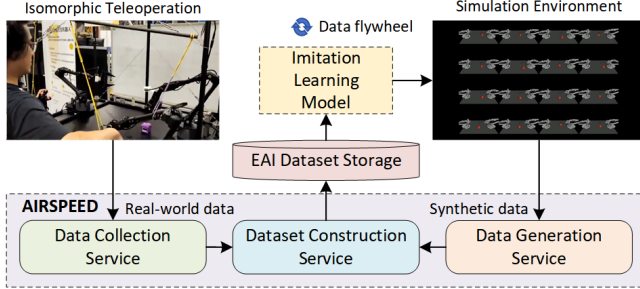
The following will illustrate, through three typical data production experiments, how AIRSPEED supports different data production methods and improves data production efficiency.

Data production for isomorphic teleoperation system. Taking ALOHA [14] as an example, Fig. 8 shows the AIRSPEED's data production process for isomorphic teleoperation. First, AIRSPEED receives real-world data from the master-slave arms and constructs the corresponding EAI dataset, which users can use to train an imitation learning model. Then, users can drive the virtual robotic arm's movements in the simulation environment using the existing imitation learning model, generating synthetic data and sending it to AIRSPEED. Finally, AIRSPEED merges the synthetic data into the EAI dataset, allowing users to train a new version of the imitation learning model with the updated integrated dataset. This process can be continuously repeated, creating a data flywheel for imitation learning.

To test the capability of AIRSPEED in accelerating data production, we conducted comparison experiments of the efficiency of using AIRSPEED in data production of isomorphic teleoperation system. As shown in Table 1, there are a total of four stages in this data production process: data collection, dataset construction (real-world data), data generation, and dataset construction (synthetic data). The experimental setup is as follows: The collection process follows the official "cube transfer" procedure of ACT [13]. The collected data includes joint and gripper motion data from four robotic arms, RGB-D data from three cameras, with a data throughput of approximately 185MB/s. The data collection service, data generation service, and dataset construction service were deployed on three separate computers, interconnected by a 1000M Ethernet.

Table 1: Comparison of the efficiency of using AIRSPEED in data production of isomorphic teleoperation system

	Data Collection (50 Samples)	Dataset Construction (Real-world Data)	Data Generation (50 Samples)	Dataset Construction (Synthetic Data)	Total
Human Operation Time Consumption	1243s	926s	187s	28s	1243s+1141s
AIRSPEED Time Consumption		26s	156s	8s	1243s+190s
Effective Acceleration Ratio	-	35.62	1.2	3.5	6.01

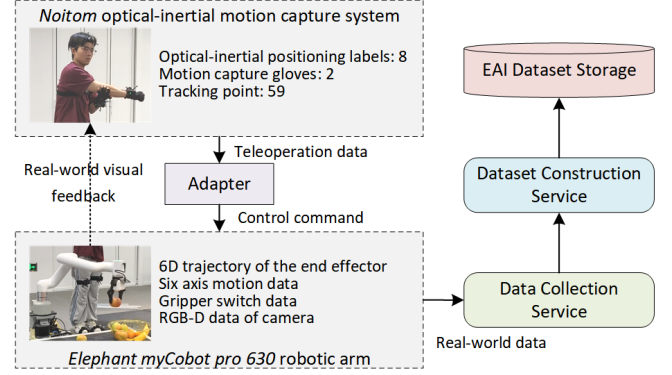
**Figure 8: AIRSPEED's data production process for isomorphic teleoperation system (taking ALOHA as an example)**

During the experiment, we recorded the average time consumed by three human operators in these four stages and calculated the effective acceleration ratio based on the use of AIRSPEED.

The results indicate that AIRSPEED significantly reduced the time consumption in the dataset construction and data generation stages. For instance, in the real-world dataset construction stage, because AIRSPEED had already begun synchronizing the construction of the dataset during the data collection stage, it could complete the dataset construction in an average of 26 seconds after the completion of data collection. In contrast, manually constructing and transferring the dataset took an average of 926 seconds. Comparatively, AIRSPEED achieved an effective acceleration ratio of up to 35.62 in the real-world dataset construction stage, and an overall effective acceleration ratio of up to 6.01. The use of AIRSPEED significantly improved the efficiency of data production in isomorphic teleoperation systems.

Data collection for optical-inertial teleoperation system.

As shown in Fig. 9, we have set up a teleoperation system for a robotic arm based on the *Noitom* optical-inertial motion capture system and the *Elephant myCobot pro 630* robotic arm. We have written a converter that transforms the teleoperation data from the operator (i.e., the 59 tracking points generated by the motion capture system) into control commands for the robotic arm. The robotic arm executes corresponding task motions based on these commands (such as picking up an apple, opening a drawer, etc.). During the teleoperation process, the operator can obtain visual feedback from the status of the robotic arm and adjust their actions in time to control the robotic arm to successfully complete the tasks. Various data generated by the robotic arm during the execution process will be collected by AIRSPEED's data collection service and organized into a dataset by the dataset construction service.

**Figure 9: AIRSPEED's data collection process for optical-inertial teleoperation system**

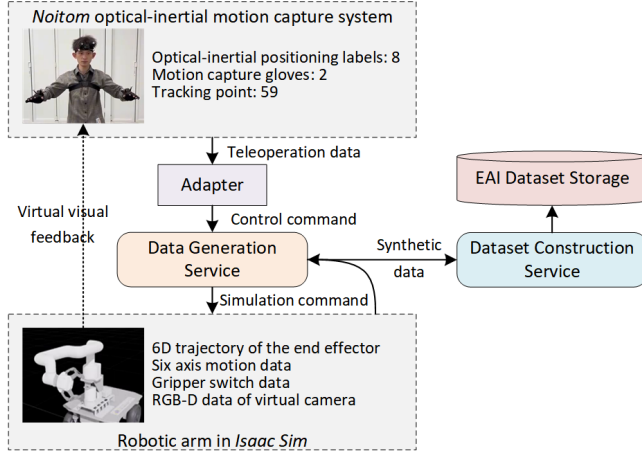
We conducted comparison experiments on the efficiency of using AIRSPEED in data collection for an optical-inertial teleoperation system. The collected data included joint and gripper motion data from one robotic arm and RGB-D data from one camera, with a data throughput of approximately 62MB/s. The rest of the experimental setup remained unchanged. As shown in Table 2, AIRSPEED significantly improved the efficiency of data production, achieving an effective acceleration ratio of up to 23.5 in the dataset construction phase.

Data generation for virtual teleoperation system. As shown in Fig. 10, users can employ the simulation interface within the data generation service to convert the operator's teleoperation data into simulation commands for the robotic arm. Consequently, the robotic arm can perform corresponding task motions within simulation environments such as Isaac Sim. During the teleoperation process, operators can obtain visual feedback through the visualization interface of the simulation platform and adjust their actions in a timely manner to control the robotic arm to successfully execute tasks. The various synthetic data generated by the robotic arm during the execution process will be returned to AIRSPEED's data generation service and organized into a dataset by the dataset construction service.

Similarly, we conducted comparison experiments on the efficiency of using AIRSPEED in the data generation process for a virtual teleoperation system. The data throughput and experimental setup remained unchanged. As shown in Table 2, AIRSPEED achieved an effective acceleration ratio of up to 7.67 in the dataset construction phase.

Table 2: Comparison of the efficiency of using AIRSPEED in data collection for optical-inertial teleoperation system and data generation process for virtual teleoperation system

	Data Collection Process for Optical-inertial Teleoperation System		Data Generation Process for Virtual Teleoperation System	
	Data Collection (20 Samples)	Dataset Construction (Real-world Data)	Data Generation (20 Samples)	Dataset Construction (Synthetic Data)
Human Operation Time Consumption	657s	517s	721s	23s
AIRSPEED Time Consumption		22s		3s
Effective Acceleration Ratio	-	23.5	-	7.67

**Figure 10: AIRSPEED's data generation process for virtual teleoperation system**

4.3 Data production performance

The data production performance of AIRSPEED is divided into two parts: the first part includes metrics related to the quality of data production, such as compression ratio and transmission latency; the second part consists of technical compatibility metrics.

Table 3 lists the metrics related to the quality of data production. For data collection, the Keyframe Selection Rate and Compression Rate can both reach up to 2%, the Compression Throughput can achieve at least 296MB/s, and the End-to-End Latency is as low as 3ms. For Data Generation, the Data Alignment Latency does not exceed a maximum of 30ms.

Table 4 demonstrates the compatibility advantages of AIRSPEED compared to other existing data collection and production platforms, devices, or software. They include PATO [22], BestMan [28], Mimicgen [29], VitRob [30], EmbodiedCity [34], MetaUrban [35], Habitat 3.0 [33], GRUtopia [36], Robo-GS [46], and ORCA [47]. It can be observed that AIRSPEED has the most comprehensive compatibility in supporting various data collection and production technologies, dataset construction, and open-source support. This showcases the advantage of AIRSPEED as a universal data production platform.

Table 3: Performance indicators of AIRSPEED

Data Collection	Keyframe Selection Rate	Up to 2%
	Compression Rate	Up to 2%
	Compression Throughput (MB/s)	At least 296MB/s
	End-to-End Latency	Minimum 3ms
Data Generation	Data Alignment Latency	Maximum 30ms

5 CONCLUSION

In this paper, we propose a universal data production open-source platform called AIRSPEED. AIRSPEED can help collect high-quality data at a low cost through hardware-software decoupling, quickly generate a large amount of data by widely docking with simulation platforms, comprehensively obtain highly generalized data by widely supporting data collection devices, and provides an automatic construction method for EAI datasets. Our experiments demonstrate the effectiveness of AIRSPEED in improving data generation efficiency and its outstanding compatibility advantages compared to other existing platforms, devices, and software. EAI is a rapidly developing field, and the questions of what data EAI model training requires, how to collect data, and how much data is needed are still subjects that require exploration. We believe that AIRSPEED not only provides a data production platform to improve data production efficiency but also can help scholars more quickly explore the answers to these questions. In future work, we will further enhance the efficiency of AIRSPEED in data collection and generation and support more collection devices and simulation platforms.

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Table 4: Data production performance of AIRSPEED

Method	Data Collection		Data Generation		Dataset Construction	Open Source
	Teleoperation	Demonstration	Data synthesis	Virtual teleoperation		
PATO	✓	✗	✓	✓	✗	✓
BestMan	✗	✗	✓	✗	✗	✓
Mimicgen	✗	✗	✗	✓	✗	✓
VitRob	✓	✗	✗	✓	✗	✗
EmbodiedCity	✗	✗	✓	✓	✓	✓
MetaUrban	✗	✗	✓	✓	✓	✓
Habitat 3.0	✗	✗	✓	✓	✓	✓
GRUtopia	✗	✗	✓	✓	✓	✓
Robo-GS	✗	✗	✓	✗	✗	✓
ORCA	✓	✗	✓	✓	✗	✗
AIRSPEED	✓	✓	✓	✓	✓	✓

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